

```
# =====
# 01_data_audit
# Airline Service Performance Visual Analytics
# =====
```

```
# 1. LOAD PACKAGES -----
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.6.1
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.6.1
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
# 2. FILE PATH -----
```

```
file_path <- "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/raw/Dataset Assignment"
```

```
# 3. CHECK SHEET NAMES -----
```

```
sheet_names <- excel_sheets(file_path)
```

```
cat("\n=== SHEET NAMES ===\n")
```

```
##
```

```
## === SHEET NAMES ===
```

```
print(sheet_names)
```

```
## [1] "2020-23 OTP" "2020"          "2019"          "2018"          "2017"
## [6] "2016"         "2015"          "2014"          "2013"          "2012"
## [11] "2011"         "2010"
```

```
# 4. READ THE MAIN SHEET -----
```

```
main_data <- read_excel(
  file_path,
  sheet = "2020-23 OTP"
)
```

Warning: Expecting numeric in M1548 / R1548C13: got 'na'

Warning: Expecting numeric in N1548 / R1548C14: got 'na'

Warning: Expecting numeric in M1550 / R1550C13: got 'na'

Warning: Expecting numeric in N1550 / R1550C14: got 'na'

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Warning: Expecting numeric in M7390 / R7390C13: got 'na'

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Warning: Expecting numeric in M7427 / R7427C13: got 'na'

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Warning: Expecting numeric in M7448 / R7448C13: got 'na'

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Warning: Expecting numeric in M7761 / R7761C13: got 'na'

Warning: Expecting numeric in N7761 / R7761C14: got 'na'

Warning: Expecting numeric in M11819 / R11819C13: got 'na'

Warning: Expecting numeric in N11819 / R11819C14: got 'na'

```
cat("\n=== ORIGINAL COLUMN NAMES ===\n")
```

```
##  
## === ORIGINAL COLUMN NAMES ===
```

```
print(names(main_data))
```

```
## [1] "Route" "Departing Port"  
## [3] "Arriving Port" "Airline"  
## [5] "Month" "Sectors Scheduled"  
## [7] "Sectors Flown" "Cancellations"  
## [9] "Departures On Time" "Arrivals On Time"  
## [11] "Departures Delayed" "Arrivals Delayed"  
## [13] "OnTime Departures \r\n(%)" "OnTime Arrivals \r\n(%)"  
## [15] "Cancellations \r\n\r\n(%)"
```

```
# 5. CREATE SIMPLE COLUMN NAMES -----
```

```
clean_column_names <- c(  
  "Route",  
  "Departing_Port",  
  "Arriving_Port",  
  "Airline",  
  "Month",  
  "Sectors_Scheduled",  
  "Sectors_Flow",  
  "Cancellations",  
  "Departures_On_Time",  
  "Arrivals_On_Time",  
  "Departures_Delayed",  
  "Arrivals_Delayed",  
  "OnTime_Departures_Percent",  
  "OnTime_Arrivals_Percent",  
  "Cancellations_Percent"  
)
```

```
names(main_data) <- clean_column_names
```

```
cat("\n=== CLEAN COLUMN NAMES ===\n")
```

```
##  
## === CLEAN COLUMN NAMES ===
```

```
print(names(main_data))
```

```
## [1] "Route" "Departing_Port"  
## [3] "Arriving_Port" "Airline"  
## [5] "Month" "Sectors_Scheduled"  
## [7] "Sectors_Flow" "Cancellations"  
## [9] "Departures_On_Time" "Arrivals_On_Time"  
## [11] "Departures_Delayed" "Arrivals_Delayed"  
## [13] "OnTime_Departures_Percent" "OnTime_Arrivals_Percent"  
## [15] "Cancellations_Percent"
```

6. CHECK EACH SHEET -----

```
sheet_summary <- data.frame(  
  Sheet = character(),  
  Rows = numeric(),  
  Columns = numeric()  
)  
  
for (sheet in sheet_names) {  
  
  data <- read_excel(  
    file_path,  
    sheet = sheet  
  )  
  
  new_row <- data.frame(  
    Sheet = sheet,  
    Rows = nrow(data),  
    Columns = ncol(data)  
  )  
  
  sheet_summary <- bind_rows(  
    sheet_summary,  
    new_row  
  )  
}
```

```
## Warning: Expecting numeric in M1548 / R1548C13: got 'na'  
  
## Warning: Expecting numeric in N1548 / R1548C14: got 'na'  
  
## Warning: Expecting numeric in M1550 / R1550C13: got 'na'  
  
## Warning: Expecting numeric in N1550 / R1550C14: got 'na'  
  
## Warning: Expecting numeric in M1554 / R1554C13: got 'na'  
  
## Warning: Expecting numeric in N1554 / R1554C14: got 'na'  
  
## Warning: Expecting numeric in M1561 / R1561C13: got 'na'  
  
## Warning: Expecting numeric in N1561 / R1561C14: got 'na'  
  
## Warning: Expecting numeric in M1624 / R1624C13: got 'na'  
  
## Warning: Expecting numeric in N1624 / R1624C14: got 'na'  
  
## Warning: Expecting numeric in M1631 / R1631C13: got 'na'  
  
## Warning: Expecting numeric in N1631 / R1631C14: got 'na'
```


Warning: Expecting numeric in M1637 / R1637C13: got 'na'

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## Warning: Expecting numeric in M2322 / R2322C13: got 'na'

## Warning: Expecting numeric in N2322 / R2322C14: got 'na'

## Warning: Expecting numeric in M3459 / R3459C13: got 'na'

## Warning: Expecting numeric in N3459 / R3459C14: got 'na'

## Warning: Expecting numeric in M3464 / R3464C13: got 'na'

## Warning: Expecting numeric in N3464 / R3464C14: got 'na'

## Warning: Expecting numeric in M3490 / R3490C13: got 'na'

## Warning: Expecting numeric in N3490 / R3490C14: got 'na'

## Warning: Expecting numeric in M3491 / R3491C13: got 'na'

## Warning: Expecting numeric in N3491 / R3491C14: got 'na'
```

```
cat("\n=== SHEET SUMMARY ===\n")
```

```
##
## === SHEET SUMMARY ===
```

```
print(sheet_summary)
```

```
##           Sheet  Rows Columns
## 1  2020-23 OTP 22037      15
## 2      2020  4197      15
## 3      2019  6409      15
## 4      2018  6548      15
## 5      2017  6470      15
## 6      2016  6342      15
## 7      2015  6174      15
## 8      2014  6045      15
## 9      2013  5722      15
## 10     2012  5236      15
## 11     2011  4944      15
## 12     2010  5050      15
```

```
# 7. CHECK 2020 DUPLICATION -----
```

```
data_2020_main <- read_excel(
  file_path,
  sheet = "2020-23 OTP"
)
```

```
## Warning: Expecting numeric in M1548 / R1548C13: got 'na'
```


Warning: Expecting numeric in N1548 / R1548C14: got 'na'

Warning: Expecting numeric in M1550 / R1550C13: got 'na'

Warning: Expecting numeric in N1550 / R1550C14: got 'na'

Warning: Expecting numeric in M1554 / R1554C13: got 'na'

Warning: Expecting numeric in N1554 / R1554C14: got 'na'

Warning: Expecting numeric in M1561 / R1561C13: got 'na'

Warning: Expecting numeric in N1561 / R1561C14: got 'na'

Warning: Expecting numeric in M1624 / R1624C13: got 'na'

Warning: Expecting numeric in N1624 / R1624C14: got 'na'

Warning: Expecting numeric in M1631 / R1631C13: got 'na'

Warning: Expecting numeric in N1631 / R1631C14: got 'na'

Warning: Expecting numeric in M1637 / R1637C13: got 'na'

Warning: Expecting numeric in N1637 / R1637C14: got 'na'

Warning: Expecting numeric in M1639 / R1639C13: got 'na'

Warning: Expecting numeric in N1639 / R1639C14: got 'na'

Warning: Expecting numeric in M1640 / R1640C13: got 'na'

Warning: Expecting numeric in N1640 / R1640C14: got 'na'

Warning: Expecting numeric in M1643 / R1643C13: got 'na'

Warning: Expecting numeric in N1643 / R1643C14: got 'na'

Warning: Expecting numeric in M1644 / R1644C13: got 'na'

Warning: Expecting numeric in N1644 / R1644C14: got 'na'

Warning: Expecting numeric in M1648 / R1648C13: got 'na'

Warning: Expecting numeric in N1648 / R1648C14: got 'na'

Warning: Expecting numeric in M1649 / R1649C13: got 'na'

Warning: Expecting numeric in N1649 / R1649C14: got 'na'

Warning: Expecting numeric in M1656 / R1656C13: got 'na'

Warning: Expecting numeric in N1656 / R1656C14: got 'na'

Warning: Expecting numeric in M1657 / R1657C13: got 'na'

Warning: Expecting numeric in N1657 / R1657C14: got 'na'

Warning: Expecting numeric in M1659 / R1659C13: got 'na'

Warning: Expecting numeric in N1659 / R1659C14: got 'na'

Warning: Expecting numeric in M1660 / R1660C13: got 'na'

Warning: Expecting numeric in N1660 / R1660C14: got 'na'

Warning: Expecting numeric in M1669 / R1669C13: got 'na'

Warning: Expecting numeric in N1669 / R1669C14: got 'na'

Warning: Expecting numeric in M1672 / R1672C13: got 'na'

Warning: Expecting numeric in N1672 / R1672C14: got 'na'

Warning: Expecting numeric in M1673 / R1673C13: got 'na'

Warning: Expecting numeric in N1673 / R1673C14: got 'na'

Warning: Expecting numeric in M1674 / R1674C13: got 'na'

Warning: Expecting numeric in N1674 / R1674C14: got 'na'

Warning: Expecting numeric in M1682 / R1682C13: got 'na'

Warning: Expecting numeric in N1682 / R1682C14: got 'na'

Warning: Expecting numeric in M1684 / R1684C13: got 'na'

Warning: Expecting numeric in N1684 / R1684C14: got 'na'

Warning: Expecting numeric in M1686 / R1686C13: got 'na'

Warning: Expecting numeric in N1686 / R1686C14: got 'na'

Warning: Expecting numeric in M1688 / R1688C13: got 'na'

Warning: Expecting numeric in N1688 / R1688C14: got 'na'

Warning: Expecting numeric in M1689 / R1689C13: got 'na'

Warning: Expecting numeric in N1689 / R1689C14: got 'na'

Warning: Expecting numeric in M1692 / R1692C13: got 'na'

Warning: Expecting numeric in N1692 / R1692C14: got 'na'

Warning: Expecting numeric in M1693 / R1693C13: got 'na'

Warning: Expecting numeric in N1693 / R1693C14: got 'na'

Warning: Expecting numeric in M2167 / R2167C13: got 'na'

Warning: Expecting numeric in N2167 / R2167C14: got 'na'

Warning: Expecting numeric in M2313 / R2313C13: got 'na'

Warning: Expecting numeric in N2313 / R2313C14: got 'na'

Warning: Expecting numeric in M2322 / R2322C13: got 'na'

Warning: Expecting numeric in N2322 / R2322C14: got 'na'

Warning: Expecting numeric in M3459 / R3459C13: got 'na'

Warning: Expecting numeric in N3459 / R3459C14: got 'na'

Warning: Expecting numeric in M3464 / R3464C13: got 'na'

Warning: Expecting numeric in N3464 / R3464C14: got 'na'

Warning: Expecting numeric in M3490 / R3490C13: got 'na'

Warning: Expecting numeric in N3490 / R3490C14: got 'na'

Warning: Expecting numeric in M3491 / R3491C13: got 'na'

Warning: Expecting numeric in N3491 / R3491C14: got 'na'

Warning: Expecting numeric in M6411 / R6411C13: got 'na'

Warning: Expecting numeric in N6411 / R6411C14: got 'na'

Warning: Expecting numeric in M6417 / R6417C13: got 'na'

Warning: Expecting numeric in N6417 / R6417C14: got 'na'

Warning: Expecting numeric in M6863 / R6863C13: got 'na'

Warning: Expecting numeric in N6863 / R6863C14: got 'na'

Warning: Expecting numeric in M6885 / R6885C13: got 'na'

Warning: Expecting numeric in N6885 / R6885C14: got 'na'

Warning: Expecting numeric in M6894 / R6894C13: got 'na'

Warning: Expecting numeric in N6894 / R6894C14: got 'na'

Warning: Expecting numeric in M6898 / R6898C13: got 'na'

Warning: Expecting numeric in N6898 / R6898C14: got 'na'

Warning: Expecting numeric in M6926 / R6926C13: got 'na'

Warning: Expecting numeric in N6926 / R6926C14: got 'na'

Warning: Expecting numeric in M6963 / R6963C13: got 'na'

Warning: Expecting numeric in N6963 / R6963C14: got 'na'

Warning: Expecting numeric in M7313 / R7313C13: got 'na'

Warning: Expecting numeric in N7313 / R7313C14: got 'na'

Warning: Expecting numeric in M7337 / R7337C13: got 'na'

Warning: Expecting numeric in N7337 / R7337C14: got 'na'

Warning: Expecting numeric in M7343 / R7343C13: got 'na'

Warning: Expecting numeric in N7343 / R7343C14: got 'na'

Warning: Expecting numeric in M7347 / R7347C13: got 'na'

Warning: Expecting numeric in N7347 / R7347C14: got 'na'

Warning: Expecting numeric in M7375 / R7375C13: got 'na'

Warning: Expecting numeric in N7375 / R7375C14: got 'na'

Warning: Expecting numeric in M7381 / R7381C13: got 'na'

Warning: Expecting numeric in N7381 / R7381C14: got 'na'

Warning: Expecting numeric in M7390 / R7390C13: got 'na'

Warning: Expecting numeric in N7390 / R7390C14: got 'na'

Warning: Expecting numeric in M7392 / R7392C13: got 'na'

Warning: Expecting numeric in N7392 / R7392C14: got 'na'

Warning: Expecting numeric in M7427 / R7427C13: got 'na'

Warning: Expecting numeric in N7427 / R7427C14: got 'na'

Warning: Expecting numeric in M7448 / R7448C13: got 'na'

Warning: Expecting numeric in N7448 / R7448C14: got 'na'

Warning: Expecting numeric in M7451 / R7451C13: got 'na'

Warning: Expecting numeric in N7451 / R7451C14: got 'na'

Warning: Expecting numeric in M7462 / R7462C13: got 'na'

Warning: Expecting numeric in N7462 / R7462C14: got 'na'

Warning: Expecting numeric in M7721 / R7721C13: got 'na'

Warning: Expecting numeric in N7721 / R7721C14: got 'na'

Warning: Expecting numeric in M7728 / R7728C13: got 'na'

Warning: Expecting numeric in N7728 / R7728C14: got 'na'

Warning: Expecting numeric in M7749 / R7749C13: got 'na'

Warning: Expecting numeric in N7749 / R7749C14: got 'na'

Warning: Expecting numeric in M7761 / R7761C13: got 'na'

Warning: Expecting numeric in N7761 / R7761C14: got 'na'

Warning: Expecting numeric in M11819 / R11819C13: got 'na'

Warning: Expecting numeric in N11819 / R11819C14: got 'na'

```
data_2020_old <- read_excel(  
  file_path,  
  sheet = "2020"  
)
```

```
## Warning: Expecting numeric in M1548 / R1548C13: got 'na'  
## Warning: Expecting numeric in N1548 / R1548C14: got 'na'  
## Warning: Expecting numeric in M1550 / R1550C13: got 'na'  
## Warning: Expecting numeric in N1550 / R1550C14: got 'na'  
## Warning: Expecting numeric in M1554 / R1554C13: got 'na'  
## Warning: Expecting numeric in N1554 / R1554C14: got 'na'  
## Warning: Expecting numeric in M1561 / R1561C13: got 'na'  
## Warning: Expecting numeric in N1561 / R1561C14: got 'na'  
## Warning: Expecting numeric in M1624 / R1624C13: got 'na'  
## Warning: Expecting numeric in N1624 / R1624C14: got 'na'  
## Warning: Expecting numeric in M1631 / R1631C13: got 'na'  
## Warning: Expecting numeric in N1631 / R1631C14: got 'na'  
## Warning: Expecting numeric in M1637 / R1637C13: got 'na'  
## Warning: Expecting numeric in N1637 / R1637C14: got 'na'  
## Warning: Expecting numeric in M1639 / R1639C13: got 'na'  
## Warning: Expecting numeric in N1639 / R1639C14: got 'na'  
## Warning: Expecting numeric in M1640 / R1640C13: got 'na'  
## Warning: Expecting numeric in N1640 / R1640C14: got 'na'  
## Warning: Expecting numeric in M1643 / R1643C13: got 'na'  
## Warning: Expecting numeric in N1643 / R1643C14: got 'na'  
## Warning: Expecting numeric in M1644 / R1644C13: got 'na'  
## Warning: Expecting numeric in N1644 / R1644C14: got 'na'
```

Warning: Expecting numeric in M1648 / R1648C13: got 'na'

Warning: Expecting numeric in N1648 / R1648C14: got 'na'

Warning: Expecting numeric in M1649 / R1649C13: got 'na'

Warning: Expecting numeric in N1649 / R1649C14: got 'na'

Warning: Expecting numeric in M1656 / R1656C13: got 'na'

Warning: Expecting numeric in N1656 / R1656C14: got 'na'

Warning: Expecting numeric in M1657 / R1657C13: got 'na'

Warning: Expecting numeric in N1657 / R1657C14: got 'na'

Warning: Expecting numeric in M1659 / R1659C13: got 'na'

Warning: Expecting numeric in N1659 / R1659C14: got 'na'

Warning: Expecting numeric in M1660 / R1660C13: got 'na'

Warning: Expecting numeric in N1660 / R1660C14: got 'na'

Warning: Expecting numeric in M1669 / R1669C13: got 'na'

Warning: Expecting numeric in N1669 / R1669C14: got 'na'

Warning: Expecting numeric in M1672 / R1672C13: got 'na'

Warning: Expecting numeric in N1672 / R1672C14: got 'na'

Warning: Expecting numeric in M1673 / R1673C13: got 'na'

Warning: Expecting numeric in N1673 / R1673C14: got 'na'

Warning: Expecting numeric in M1674 / R1674C13: got 'na'

Warning: Expecting numeric in N1674 / R1674C14: got 'na'

Warning: Expecting numeric in M1682 / R1682C13: got 'na'

Warning: Expecting numeric in N1682 / R1682C14: got 'na'

Warning: Expecting numeric in M1684 / R1684C13: got 'na'

Warning: Expecting numeric in N1684 / R1684C14: got 'na'

Warning: Expecting numeric in M1686 / R1686C13: got 'na'

Warning: Expecting numeric in N1686 / R1686C14: got 'na'

Warning: Expecting numeric in M1688 / R1688C13: got 'na'

Warning: Expecting numeric in N1688 / R1688C14: got 'na'

Warning: Expecting numeric in M1689 / R1689C13: got 'na'

Warning: Expecting numeric in N1689 / R1689C14: got 'na'

Warning: Expecting numeric in M1692 / R1692C13: got 'na'

Warning: Expecting numeric in N1692 / R1692C14: got 'na'

Warning: Expecting numeric in M1693 / R1693C13: got 'na'

Warning: Expecting numeric in N1693 / R1693C14: got 'na'

Warning: Expecting numeric in M2167 / R2167C13: got 'na'

Warning: Expecting numeric in N2167 / R2167C14: got 'na'

Warning: Expecting numeric in M2313 / R2313C13: got 'na'

Warning: Expecting numeric in N2313 / R2313C14: got 'na'

Warning: Expecting numeric in M2322 / R2322C13: got 'na'

Warning: Expecting numeric in N2322 / R2322C14: got 'na'

Warning: Expecting numeric in M3459 / R3459C13: got 'na'

Warning: Expecting numeric in N3459 / R3459C14: got 'na'

Warning: Expecting numeric in M3464 / R3464C13: got 'na'

Warning: Expecting numeric in N3464 / R3464C14: got 'na'

Warning: Expecting numeric in M3490 / R3490C13: got 'na'

Warning: Expecting numeric in N3490 / R3490C14: got 'na'

Warning: Expecting numeric in M3491 / R3491C13: got 'na'

Warning: Expecting numeric in N3491 / R3491C14: got 'na'


```

names(data_2020_main) <- clean_column_names
names(data_2020_old) <- clean_column_names

data_2020_main <- data_2020_main %>%
  filter(
    format(as.Date(Month), "%Y") == "2020"
  )

duplicate_2020 <- inner_join(
  data_2020_main,
  data_2020_old,
  by = c(
    "Route",
    "Departing_Port",
    "Arriving_Port",
    "Airline",
    "Month"
  )
)

cat("\n=== 2020 DUPLICATE CHECK ===\n")

```

```

##
## === 2020 DUPLICATE CHECK ===

```

```

cat(
  "2020 rows in 2020-23 OTP:",
  nrow(data_2020_main),
  "\n"
)

```

```

## 2020 rows in 2020-23 OTP: 4197

```

```

cat(
  "Rows in standalone 2020 sheet:",
  nrow(data_2020_old),
  "\n"
)

```

```

## Rows in standalone 2020 sheet: 4197

```

```

cat(
  "Matching 2020 keys:",
  nrow(duplicate_2020),
  "\n"
)

```

```

## Matching 2020 keys: 4197

```

```
# 8. CHOOSE SHEETS FOR THE MASTER DATA -----
```

```
sheets_to_use <- sheet_names[
  sheet_names != "2020"
]
```

```
cat("\n=== SHEETS USED ===\n")
```

```
##
```

```
## === SHEETS USED ===
```

```
print(sheets_to_use)
```

```
## [1] "2020-23 OTP" "2019"          "2018"          "2017"          "2016"
## [6] "2015"          "2014"          "2013"          "2012"          "2011"
## [11] "2010"
```

```
# 9. READ AND COMBINE THE SHEETS -----
```

```
data_list <- list()
```

```
for (sheet in sheets_to_use) {
```

```
  data <- read_excel(
    file_path,
    sheet = sheet,
    range = cell_cols("A:O")
  )
```

```
  names(data) <- clean_column_names
```

```
  data$Source_Sheet <- sheet
```

```
  data_list[[sheet]] <- data
}
```

```
## Warning: Expecting numeric in M1548 / R1548C13: got 'na'
```

```
## Warning: Expecting numeric in N1548 / R1548C14: got 'na'
```

```
## Warning: Expecting numeric in M1550 / R1550C13: got 'na'
```

```
## Warning: Expecting numeric in N1550 / R1550C14: got 'na'
```

```
## Warning: Expecting numeric in M1554 / R1554C13: got 'na'
```

```
## Warning: Expecting numeric in N1554 / R1554C14: got 'na'
```

```
## Warning: Expecting numeric in M1561 / R1561C13: got 'na'
```

Warning: Expecting numeric in N1561 / R1561C14: got 'na'

Warning: Expecting numeric in M1624 / R1624C13: got 'na'

Warning: Expecting numeric in N1624 / R1624C14: got 'na'

Warning: Expecting numeric in M1631 / R1631C13: got 'na'

Warning: Expecting numeric in N1631 / R1631C14: got 'na'

Warning: Expecting numeric in M1637 / R1637C13: got 'na'

Warning: Expecting numeric in N1637 / R1637C14: got 'na'

Warning: Expecting numeric in M1639 / R1639C13: got 'na'

Warning: Expecting numeric in N1639 / R1639C14: got 'na'

Warning: Expecting numeric in M1640 / R1640C13: got 'na'

Warning: Expecting numeric in N1640 / R1640C14: got 'na'

Warning: Expecting numeric in M1643 / R1643C13: got 'na'

Warning: Expecting numeric in N1643 / R1643C14: got 'na'

Warning: Expecting numeric in M1644 / R1644C13: got 'na'

Warning: Expecting numeric in N1644 / R1644C14: got 'na'

Warning: Expecting numeric in M1648 / R1648C13: got 'na'

Warning: Expecting numeric in N1648 / R1648C14: got 'na'

Warning: Expecting numeric in M1649 / R1649C13: got 'na'

Warning: Expecting numeric in N1649 / R1649C14: got 'na'

Warning: Expecting numeric in M1656 / R1656C13: got 'na'

Warning: Expecting numeric in N1656 / R1656C14: got 'na'

Warning: Expecting numeric in M1657 / R1657C13: got 'na'

Warning: Expecting numeric in N1657 / R1657C14: got 'na'

Warning: Expecting numeric in M1659 / R1659C13: got 'na'

Warning: Expecting numeric in N1659 / R1659C14: got 'na'

Warning: Expecting numeric in M1660 / R1660C13: got 'na'

Warning: Expecting numeric in N1660 / R1660C14: got 'na'

Warning: Expecting numeric in M1669 / R1669C13: got 'na'

Warning: Expecting numeric in N1669 / R1669C14: got 'na'

Warning: Expecting numeric in M1672 / R1672C13: got 'na'

Warning: Expecting numeric in N1672 / R1672C14: got 'na'

Warning: Expecting numeric in M1673 / R1673C13: got 'na'

Warning: Expecting numeric in N1673 / R1673C14: got 'na'

Warning: Expecting numeric in M1674 / R1674C13: got 'na'

Warning: Expecting numeric in N1674 / R1674C14: got 'na'

Warning: Expecting numeric in M1682 / R1682C13: got 'na'

Warning: Expecting numeric in N1682 / R1682C14: got 'na'

Warning: Expecting numeric in M1684 / R1684C13: got 'na'

Warning: Expecting numeric in N1684 / R1684C14: got 'na'

Warning: Expecting numeric in M1686 / R1686C13: got 'na'

Warning: Expecting numeric in N1686 / R1686C14: got 'na'

Warning: Expecting numeric in M1688 / R1688C13: got 'na'

Warning: Expecting numeric in N1688 / R1688C14: got 'na'

Warning: Expecting numeric in M1689 / R1689C13: got 'na'

Warning: Expecting numeric in N1689 / R1689C14: got 'na'

Warning: Expecting numeric in M1692 / R1692C13: got 'na'

Warning: Expecting numeric in N1692 / R1692C14: got 'na'

Warning: Expecting numeric in M1693 / R1693C13: got 'na'

Warning: Expecting numeric in N1693 / R1693C14: got 'na'

Warning: Expecting numeric in M2167 / R2167C13: got 'na'

Warning: Expecting numeric in N2167 / R2167C14: got 'na'

Warning: Expecting numeric in M2313 / R2313C13: got 'na'

Warning: Expecting numeric in N2313 / R2313C14: got 'na'

Warning: Expecting numeric in M2322 / R2322C13: got 'na'

Warning: Expecting numeric in N2322 / R2322C14: got 'na'

Warning: Expecting numeric in M3459 / R3459C13: got 'na'

Warning: Expecting numeric in N3459 / R3459C14: got 'na'

Warning: Expecting numeric in M3464 / R3464C13: got 'na'

Warning: Expecting numeric in N3464 / R3464C14: got 'na'

Warning: Expecting numeric in M3490 / R3490C13: got 'na'

Warning: Expecting numeric in N3490 / R3490C14: got 'na'

Warning: Expecting numeric in M3491 / R3491C13: got 'na'

Warning: Expecting numeric in N3491 / R3491C14: got 'na'

Warning: Expecting numeric in M6411 / R6411C13: got 'na'

Warning: Expecting numeric in N6411 / R6411C14: got 'na'

Warning: Expecting numeric in M6417 / R6417C13: got 'na'

Warning: Expecting numeric in N6417 / R6417C14: got 'na'

Warning: Expecting numeric in M6863 / R6863C13: got 'na'

Warning: Expecting numeric in N6863 / R6863C14: got 'na'

Warning: Expecting numeric in M6885 / R6885C13: got 'na'

Warning: Expecting numeric in N6885 / R6885C14: got 'na'

Warning: Expecting numeric in M6894 / R6894C13: got 'na'

Warning: Expecting numeric in N6894 / R6894C14: got 'na'

Warning: Expecting numeric in M6898 / R6898C13: got 'na'

Warning: Expecting numeric in N6898 / R6898C14: got 'na'

Warning: Expecting numeric in M6926 / R6926C13: got 'na'

Warning: Expecting numeric in N6926 / R6926C14: got 'na'

Warning: Expecting numeric in M6963 / R6963C13: got 'na'

Warning: Expecting numeric in N6963 / R6963C14: got 'na'

Warning: Expecting numeric in M7313 / R7313C13: got 'na'

Warning: Expecting numeric in N7313 / R7313C14: got 'na'

Warning: Expecting numeric in M7337 / R7337C13: got 'na'

Warning: Expecting numeric in N7337 / R7337C14: got 'na'

Warning: Expecting numeric in M7343 / R7343C13: got 'na'

Warning: Expecting numeric in N7343 / R7343C14: got 'na'

Warning: Expecting numeric in M7347 / R7347C13: got 'na'

Warning: Expecting numeric in N7347 / R7347C14: got 'na'

Warning: Expecting numeric in M7375 / R7375C13: got 'na'

Warning: Expecting numeric in N7375 / R7375C14: got 'na'

Warning: Expecting numeric in M7381 / R7381C13: got 'na'

Warning: Expecting numeric in N7381 / R7381C14: got 'na'

Warning: Expecting numeric in M7390 / R7390C13: got 'na'

Warning: Expecting numeric in N7390 / R7390C14: got 'na'

Warning: Expecting numeric in M7392 / R7392C13: got 'na'

Warning: Expecting numeric in N7392 / R7392C14: got 'na'

Warning: Expecting numeric in M7427 / R7427C13: got 'na'

```
## Warning: Expecting numeric in N7427 / R7427C14: got 'na'
## Warning: Expecting numeric in M7448 / R7448C13: got 'na'
## Warning: Expecting numeric in N7448 / R7448C14: got 'na'
## Warning: Expecting numeric in M7451 / R7451C13: got 'na'
## Warning: Expecting numeric in N7451 / R7451C14: got 'na'
## Warning: Expecting numeric in M7462 / R7462C13: got 'na'
## Warning: Expecting numeric in N7462 / R7462C14: got 'na'
## Warning: Expecting numeric in M7721 / R7721C13: got 'na'
## Warning: Expecting numeric in N7721 / R7721C14: got 'na'
## Warning: Expecting numeric in M7728 / R7728C13: got 'na'
## Warning: Expecting numeric in N7728 / R7728C14: got 'na'
## Warning: Expecting numeric in M7749 / R7749C13: got 'na'
## Warning: Expecting numeric in N7749 / R7749C14: got 'na'
## Warning: Expecting numeric in M7761 / R7761C13: got 'na'
## Warning: Expecting numeric in N7761 / R7761C14: got 'na'
## Warning: Expecting numeric in M11819 / R11819C13: got 'na'
## Warning: Expecting numeric in N11819 / R11819C14: got 'na'
```

```
master_raw <- bind_rows(data_list)
```

```
# 10. KEEP REAL DATA ROWS -----
```

```
master_data <- master_raw %>%
  filter(
    !is.na(Route),
    Route != "",
    !is.na(Month)
  )
```

```
# 11. CHECK DATASET SIZE -----
```

```
cat("\n=== DATASET OVERVIEW ===\n")
```

```
##
## === DATASET OVERVIEW ===
```

```
cat(  
  "Number of rows:",  
  nrow(master_data),  
  "\n"  
)
```

Number of rows: 80972

```
cat(  
  "Number of columns:",  
  ncol(master_data),  
  "\n"  
)
```

Number of columns: 16

```
cat(  
  "Date range:",  
  as.character(min(master_data$Month)),  
  "to",  
  as.character(max(master_data$Month)),  
  "\n"  
)
```

Date range: 2010-01-01 to 2024-02-01

```
cat(  
  "Number of months:",  
  n_distinct(master_data$Month),  
  "\n"  
)
```

Number of months: 170

```
cat(  
  "Number of airline labels:",  
  n_distinct(master_data$Airline),  
  "\n"  
)
```

Number of airline labels: 14

```
cat(  
  "Number of route labels:",  
  n_distinct(master_data$Route),  
  "\n"  
)
```

Number of route labels: 155


```
# 12. CHECK AIRPORT NAMES -----
```

```
airport_names <- c(
  master_data$Departing_Port,
  master_data$Arriving_Port
)

airport_names <- unique(airport_names)

airport_names <- airport_names[
  !is.na(airport_names) &
  airport_names != "All Ports"
]

cat("\n=== AIRPORT CHECK ===\n")
```

```
##
## === AIRPORT CHECK ===
```

```
cat(
  "Number of physical airport names:",
  length(airport_names),
  "\n"
)
```

```
## Number of physical airport names: 42
```

```
print(
  sort(airport_names)
)
```

```
## [1] "Adelaide"      "Albury"        "Alice Springs" "Armidale"
## [5] "Ayers Rock"    "Ballina"       "Brisbane"     "Broome"
## [9] "Bundaberg"     "Cairns"        "Canberra"     "Coffs Harbour"
## [13] "Darwin"        "Devonport"     "Dubbo"        "Emerald"
## [17] "Geraldton"     "Gladstone"     "Gold Coast"   "Hamilton Island"
## [21] "Hobart"        "Kalgoorlie"    "Karratha"     "Launceston"
## [25] "Mackay"        "Melbourne"     "Mildura"      "Moranbah"
## [29] "Mount Isa"     "Newcastle"     "Newman"       "Perth"
## [33] "Port Hedland"  "Port Lincoln"  "Port Macquarie" "Proserpine"
## [37] "Rockhampton"   "Sunshine Coast" "Sydney"       "Tamworth"
## [41] "Townsville"    "Wagga Wagga"
```

```
# 13. CHECK AIRLINE NAMES -----
```

```
cat("\n=== AIRLINE LABELS ===\n")
```

```
##
## === AIRLINE LABELS ===
```

```

print(
  sort(
    unique(master_data$Airline)
  )
)

## [1] "All Airlines"
## [2] "Bonza"
## [3] "Jetstar"
## [4] "Qantas"
## [5] "QantasLink"
## [6] "Regional Express"
## [7] "Rex Airlines"
## [8] "Skytrans"
## [9] "Skywest"
## [10] "Tigerair Australia"
## [11] "virgin Australia"
## [12] "Virgin Australia"
## [13] "Virgin Australia - ATR/F100 Operations"
## [14] "Virgin Australia Regional Airlines"

```

```

# 14. CREATE FOUR ANALYTICAL LEVELS -----

```

```

network_month <- master_data %>%
  filter(
    Route == "All Ports-All Ports",
    Airline == "All Airlines"
  )

airline_month <- master_data %>%
  filter(
    Route == "All Ports-All Ports",
    Airline != "All Airlines"
  )

route_month <- master_data %>%
  filter(
    Route != "All Ports-All Ports",
    Airline == "All Airlines"
  )

airline_route_month <- master_data %>%
  filter(
    Route != "All Ports-All Ports",
    Airline != "All Airlines"
  )

cat("\n=== ANALYTICAL DATA LEVELS ===\n")

```

```

##

```

```
## === ANALYTICAL DATA LEVELS ===
```

```
cat(  
  "Network-month rows:",  
  nrow(network_month),  
  "\n"  
)
```

```
## Network-month rows: 170
```

```
cat(  
  "Airline-month rows:",  
  nrow(airline_month),  
  "\n"  
)
```

```
## Airline-month rows: 1174
```

```
cat(  
  "Route-month rows:",  
  nrow(route_month),  
  "\n"  
)
```

```
## Route-month rows: 20156
```

```
cat(  
  "Airline-route-month rows:",  
  nrow(airline_route_month),  
  "\n"  
)
```

```
## Airline-route-month rows: 59472
```

```
cat(  
  "Total rows across four levels:",  
  nrow(network_month) +  
  nrow(airline_month) +  
  nrow(route_month) +  
  nrow(airline_route_month),  
  "\n"  
)
```

```
## Total rows across four levels: 80972
```

```
# 15. CHECK MISSING OTP VALUES -----
```

```
departure_otp <- as.character(  
  master_data$OnTime_Departures_Percent  
)
```

```

arrival_otp <- as.character(
  master_data$OnTime_Arrivals_Percent
)

missing_departure_otp <- sum(
  is.na(departure_otp) |
  tolower(trimws(departure_otp)) == "na"
)

missing_arrival_otp <- sum(
  is.na(arrival_otp) |
  tolower(trimws(arrival_otp)) == "na"
)

cat("\n=== MISSING OTP CHECK ===\n")

```

```

##
## === MISSING OTP CHECK ===

```

```

cat(
  "Missing Departure OTP values:",
  missing_departure_otp,
  "\n"
)

```

```

## Missing Departure OTP values: 91

```

```

cat(
  "Missing Arrival OTP values:",
  missing_arrival_otp,
  "\n"
)

```

```

## Missing Arrival OTP values: 91

```

```

# 16. CHECK ZERO SECTORS FLOWN -----

```

```

zero_flown <- master_data %>%
  filter(
    Sectors_Flown == 0
  )

cat("\n=== ZERO SECTORS FLOWN ===\n")

```

```

##
## === ZERO SECTORS FLOWN ===

```

```

cat(
  "Rows with zero sectors flown:",
  nrow(zero_flown),
  "\n"
)

```

```
## Rows with zero sectors flown: 91
```

```
print(  
  zero_flown %>%  
    select(  
      Route,  
      Departing_Port,  
      Arriving_Port,  
      Airline,  
      Month,  
      Sectors_Scheduled,  
      Sectors_Flown,  
      Cancellations,  
      OnTime_Departures_Percent,  
      OnTime_Arrivals_Percent  
    ) %>%  
    head(10)  
)
```

```
## # A tibble: 10 x 10  
##   Route Departing_Port Arriving_Port Airline Month  
##   <chr> <chr> <chr> <chr> <dtm>  
## 1 Canberra-Melbourne Canberra Melbourne Qantas 2020-04-01 00:00:00  
## 2 Kalgoorlie-Perth Kalgoorlie Perth Qantas 2020-04-01 00:00:00  
## 3 Melbourne-Canberra Melbourne Canberra Qantas 2020-04-01 00:00:00  
## 4 Perth-Kalgoorlie Perth Kalgoorlie Qantas 2020-04-01 00:00:00  
## 5 Armidale-Sydney Armidale Sydney Regional~ 2020-04-01 00:00:00  
## 6 Sydney-Armidale Sydney Armidale Regional~ 2020-04-01 00:00:00  
## 7 Adelaide-Brisbane Adelaide Brisbane Virgin A~ 2020-04-01 00:00:00  
## 8 Adelaide-Sydney Adelaide Sydney Virgin A~ 2020-04-01 00:00:00  
## 9 Brisbane-Adelaide Brisbane Adelaide Virgin A~ 2020-04-01 00:00:00  
## 10 Brisbane-Canberra Brisbane Canberra Virgin A~ 2020-04-01 00:00:00  
## # i 5 more variables: Sectors_Scheduled <dbl>, Sectors_Flown <dbl>,  
## # Cancellations <dbl>, OnTime_Departures_Percent <dbl>,  
## # OnTime_Arrivals_Percent <dbl>
```

```
# 17. CHECK DUPLICATED KEYS -----
```

```
duplicate_keys <- master_data %>%  
  count(  
    Route,  
    Departing_Port,  
    Arriving_Port,  
    Airline,  
    Month  
  ) %>%  
  filter(  
    n > 1  
  )  
  
cat("\n=== DUPLICATE KEY CHECK ===\n")
```

```
##  
## === DUPLICATE KEY CHECK ===
```

```
cat(
  "Number of duplicated keys:",
  nrow(duplicate_keys),
  "\n"
)
```

```
## Number of duplicated keys: 0
```

```
print(
  head(
    duplicate_keys,
    10
  )
)
```

```
## # A tibble: 0 x 6
## # i 6 variables: Route <chr>, Departing_Port <chr>, Arriving_Port <chr>,
## #   Airline <chr>, Month <dtm>, n <int>
```

```
# 18. AUDIT FINISHED -----
```

```
cat("\n=== AUDIT FINISHED ===\n")
```

```
##
## === AUDIT FINISHED ===
```